

Screen Augmentation Technique Using AR Glasses and Smartphone without External Sensors

Kazuhiro Takeda
al20096@shibaura-it.ac.jp
Shibaura Institute of Technology
Tokyo, Japan

Hiroyuki Manabe
manabehiroyuki@acm.org
Shibaura Institute of Technology
Tokyo, Japan

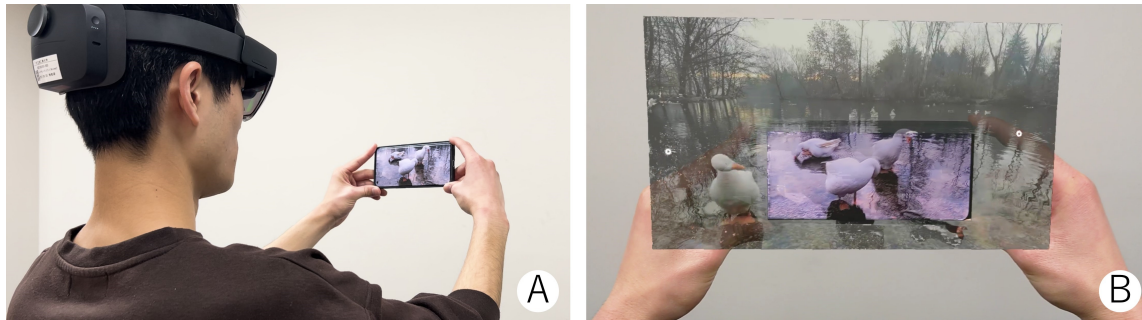


Figure 1: Smartphone holding posture wearing AR glasses (A), and how the user sees the combined images (B).

ABSTRACT

Screen augmentation using a smartphone with AR glasses can enhance visual experiences, and many studies on the combination have been conducted. Tracking the smartphone is the key to seamless augmentation; however, existing techniques suffer limited availability and accuracy. We propose a technique of screen augmentation with AR glasses and smartphone that does not require external sensors. It is assumed that some tracking error exists, and video processing is performed to make it difficult for the user to perceive any difference between the two screens. It is confirmed that our technique makes the difference less perceptible. Several applications are implemented to explore the interactions achieved with the technique.

CCS CONCEPTS

• Human-centered computing → Mixed / augmented reality.

KEYWORDS

Augmented reality, device tracking, boundaries

ACM Reference Format:

Kazuhiro Takeda and Hiroyuki Manabe. 2024. Screen Augmentation Technique Using AR Glasses and Smartphone without External Sensors. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3613905.3648664>

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3648664>

1 INTRODUCTION

Augmented reality (AR) technology continues to advance, and we can currently enjoy AR experiences using smartphones or other devices. Examples of AR applications include simulating the installation of furniture before purchase, such as IKEA Place, and experiencing virtual make-up sessions and try-before-buy fashion. An example of AR is augmented reality television (ARTV), which adds new topical information to the content of the TV screen [11, 18, 21]. This combines two or more screens and provides a new viewing experience for the user. Instead of augmenting fixed located screens, research is advancing towards the augmentation of the mobile device's screen using AR glasses, head-mounted devices with an optical see-through display.

When augmenting mobile screens, accurate tracking is the key to seamless augmentation. This is because the user can easily perceive the gap between the screens if the tracking error is not addressed. Most studies addressing this kind of augmentation uses an external motion capture system. While this is effective when exploring the interaction itself, the user experience is limited to the setup environment and availability is limited. Some proposals do not require external sensors. Although they allow the user to experience augmentation anytime anywhere, they are troubled by the issues of low tracking accuracy and drift. Ideally, achieving both high tracking accuracy and high availability is necessary.

The technique proposed in this paper augments the combination of smartphone with AR glasses, anytime, anywhere. The contributions of this paper include (1) implement and evaluate a simple but robust smartphone tracking method without using external sensors, (2) propose an image processing method that makes the screen differences less perceptible and conduct a user study, and (3) develop several applications to explore the design space.

2 RELATED WORK

There are several approaches to enhance the AR screen by combining two or more devices, for example, augmenting screens in fixed location such as monitors or rooms, as well as augmenting mobile screens. The latter needs to continuously track the screen.

2.1 Augmenting screens in fixed location

The CAVE, a cube that surrounds the user with a contiguous display surface, provides a different VR experience than using a HMD [5]. Techniques have also been proposed to bring similar ideas to the home. For example, Illumiroom expands the gaming experience by using a projector to create peripheral illusions around the monitor [10]. RoomAlive uses several projector-depth camera units to transform a room into an immersive gaming experience [9]. Visual experiences through a HMD, augmented monitor using projectors, and a monitor have been compared [14]. ARTV expands the TV in the living room and extends it via a tablet [21] and an AR HMD [11, 18] has been examined. Since they augment statically located screens, seamless expansion is possible once careful prior alignment is achieved. However, they are available only in the setup environment.

Several techniques can enhance mobile displays with AR HMDs but display positions are not tracked. For example, an AR HMD enhances the reading experience on smartphones [6], and annotations are set in halo around the tablet using an AR HMD [1]. It works if the position of the mobile display can be assumed to be fixed, or the location of augmented information shown is not important.

2.2 Augmenting mobile screens

Augmenting mobile screens, such as a smartphone or smartwatch, eases their limited display area. Tracking the mobile display and how to show the augmented screen is the key to seamlessly augmenting the mobile display, otherwise there could be a gap between the mobile display and the augmentation screen. Motion capturing systems, such as OptiTrack and Vicon, provide precise tracking accuracy and are often used. For example, HoloBar [19], MARVIS [12], SecondSight [17], BISHARE [22], and MultiFi [7] use AR HMDs to augment smartphones or smartwatch displays with tracking provided by a motion capture system. Video-see through HMDs, referred to as VR HMDs, provide a generally wider field of view compared with AR HMDs and can overwrite the mobile display to allow seamless augmentation even if there is tracking error. Some studies have used VR HMDs to evaluate the extension [8, 16]. The use of external motion capture systems or VR HMDs can help in understanding interactions, but the user can receive the benefit only in the setup environment.

Some studies have been done without using external sensors for device tracking. For example, some studies attach visual markers to the mobile device that are then captured by the AR HMD [2, 4, 20]. TrackCap attaches a marker to a cap and reads it from the smartphone [15]. PAIR [20] also runs SLAM [3] on both the mobile device and the HMD. Screen tracking via a hand tracking function on the VR HMD without visual markers has been examined [13], however, it used a transparent board instead of a smartphone. Although these techniques allow the user to experience the augmentation

anytime anywhere, they often suffer low tracking accuracy and poor appearance due to visual markers.

3 PROPOSED TECHNIQUE

As mentioned below, the combination of a mobile device and AR glasses makes it difficult to achieve both high tracking accuracy and high availability in screen augmentation. This paper proposes a smartphone screen augmentation technique that uses only AR glasses and a smartphone, and no external sensors are needed. We start by accepting that the position of the smartphone is not known accurately and aim to add an effect to the image so that the user does not feel any significant displacement between the screen and the augmented information. The image is divided into two parts, one for AR glasses and the other for the smartphone, and the areas are continuously updated according to the tracking results. This not only makes it possible to view augmented images anytime, anywhere, but also allows the user to view arbitrary areas of interest at high resolution.

3.1 Implementation

A prototype was created out using Microsoft's HoloLens 2 as the AR glasses and an OPPO Find X3 Pro as the smartphone. Several methods are possible for tracking the smartphone, but smartphone holding hand posture is predefined in this implementation and the HoloLens 2 built-in hand tracking function is used. Thanks to recent advances in technology, the built-in hand tracking is robust and easy to use. Note that the use of hand tracking built into HMDs for smartphone tracking was considered in [13], however, this study tests an actual smartphone. The way the phone is held is predefined as shown in Figure 1 (A), the four corners of the smartphone are held with the thumbs and index fingers of both hands. The smartphone is estimated to be the rectangle defined by the four fingers. In our current implementation, smartphone rotation is not supported. The estimated position of the smartphone is sent to the server using Websocket, and the image on the AR glasses masks out the smartphone's area. The smartphone displays the image that matches its position, so the user sees the combined result of the smartphone display and the AR image.

There is some error in hand tracking that can cause gaps or overlaps between the two images. Three methods were implemented to make the differences less perceptible, (A) the mask is made larger than the smartphone screen which emulates the bezels of the traditional display, (B) the mask is made smaller than the smartphone screen which yields aggressive overlap of the two images, and (C) the mask is smaller than the smartphone screen and the overlapping parts of the AR image are slightly blurred. As a reference, (D) made the mask match the size of the smartphone. The four methods are shown in Figure 2.

Videos were played on HoloLens 2 and the smartphone a simultaneously after offsetting the slight time difference between the two display systems.

Materials for the AR glasses and the smartphone were created in Unity and the images were processed on a PC for display on HoloLens 2 using the Holographic Remoting Player. Figure 1 (B) shows a scene of what the user sees using this prototype.

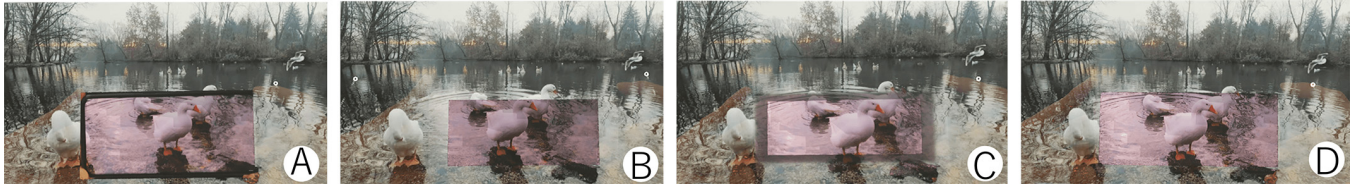


Figure 2: Four methods of setting AR mask, (A) the mask is larger than the smartphone screen, (B) the mask is smaller, (C) AR glasses blur the area where the two images overlap in addition to (B), and (D) the size of the phone screen and the mask are the same.

4 EVALUATIONS

Two evaluations were conducted; one measured the spatial differences between the two images, the other compared the four methods in terms of user perceived differences.

4.1 Evaluations of spatial misalignment

Several factors cause spatial misalignment between the two images. For example, tracking error when the smartphone is stationary and tracking delay when the smartphone is moving.

The experiment was conducted as follows. A full HD image with horizontal and vertical lines drawn on a 10 pixel grid was shown at a distance of approximately 0.5 m from the eye position. The display resolution of one eye of HoloLens 2 is 2K, the viewing angle is approximately 52° diagonally, and the display resolution of the smartphone is QHD+ at 6.7 inch. The smartphone was placed stationary at the center of the image and the author watched the image wearing the AR glasses. There were both horizontal and vertical gaps between the lines in the two images and the gaps were visually measured in units of AR pixels. The horizontal gap was a maximum of about 15 to 20 px and vertical about 10 px, which correspond to almost 1% against the whole image. Then, the misalignment created by moving the smartphone was measured using the video recorded by the AR glasses showing the smartphone's screen and the AR display. The video was inspected frame by frame and the positions of the smartphone, mask, and image displayed on the smartphone were measured. Note that the video is not the same as the user's view because the video was captured using a built-in camera whose position differed from the user's eye. That means that the three positions were recorded as relative values. The results of moving the smartphone at speeds of approximately 6.6 cm/s and 10 cm/s are shown in Figure 3. There is a delay of approximately 10 frames from the time the smartphone starts to move until the movement is reflected in the AR image. When the smartphone is moving rapidly, the overshoot takes up about 15 frames. These results confirm that there are differences between the two images, and they become larger when the smartphone is moving.

4.2 User study on how the difference is perceived

Five videos were prepared to compare the four methods, (A) larger mask, (B) smaller mask, (C) smaller mask + blurring, (D) mask of the same size, to confirm the degree of perceptible difference. The five videos differed in terms of motion and brightness. Five participants

were asked to watch a video using the four different methods, in total 5 videos x 4 methods x 5 participants. After viewing each video, a questionnaire was given, in which participants were asked to judge the absolute and relative difficulty in perceiving the video boundary. The experiment lasted approximately one hour per person, and each subject was rewarded with an Amazon gift card of 1,000 Japanese yen. In order to reduce the influence of order, the method and the videos were randomized for each subject. The brightness of the AR glasses and of the smartphone were adjusted for each video before starting the experiment. Each method was shown for 25 seconds, and after viewing the four methods, the subjects could freely switch between methods for a better and comparison before evaluating them.

Figure 4 shows the average of the rankings and all the answers for each video. It shows that the blurring method (C) provides the best result, the least perceptible difference, for all videos. Actually, most of the answers for method (C) were 5 on the 5-level Likert scale used. Note that the higher score, the better the result in this scale, and a score of 5 means that the participant hardly perceived the differences. In addition, method (B) with a smaller mask yielded less or the equal degree of perceptible difference as method (D) as the reference, but the result did depend on the video. On the other hand, method (A) with a larger mask yielded the most perceptible displacement. Subjects commented, "the difference between the methods is smaller for videos in which the object does not move much and in which the background is dark," and "when the background is dark, the difference between the methods is less noticeable with method (A)." Some participants commented the overlapped part of the images looked three-dimensional, while others felt that it was out of alignment.

5 DISCUSSION

The prototype was found to have several limitations and possible improvements. We discuss these in addition to applications developed to explore the design space.

5.1 Limitations

Although it was confirmed that the current prototype achieved a less perceptible difference between the two images, it is highly desired to achieve more accurate tracking with less delay. Possible improvements include, for example, use of the depth information of the smartphone surface as captured by the camera embedded in the AR glasses, the use of the camera or IMU sensors in the

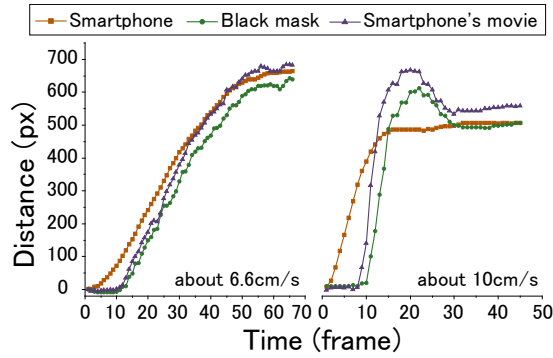


Figure 3: Tracking gap result: (left) when the phone is moved at about 6.6 cm/s. (right) when the phone is moved at about 10 cm/s. There is a delay of about 10 frames after moving the phone, and overshoot occurs for about 15 frames when the speed is high.

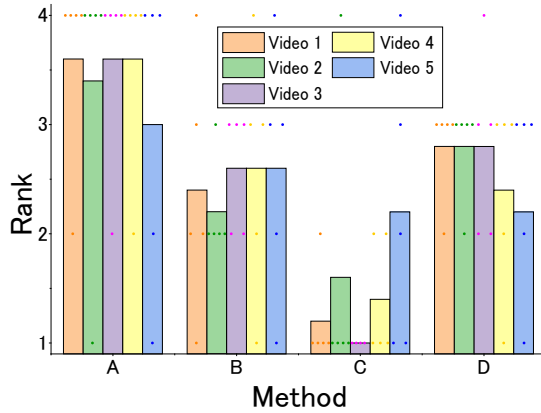


Figure 4: Average rank and all participants' answers of the difficulty in perceiving the video boundary. The bar graph shows the average rank and participants' answers are indicated by dots. The lower the number, the better the result. The blurring method (C) was the least perceptible to the displacement for all the videos.

smartphone, and the use of SLAM technology as employed in conventional research. It is expected these improvements could prevent the overshoot shown in Figure 3.

While method (C) achieved the best performance in the experiment, there could be more effective methods. For example, a gradient in brightness in addition to blurring, and a method that adapted to the content could yield better results.

The brightness difference of the two images is also a big factor in perceiving spatial gaps between the images. In the current configuration, the brightness adjustment was conducted manually according to the content and the user. It is necessary to eliminate this manual procedure.



Figure 5: Developed applications: (A) a fish photography game, (B) a cross-sectional display application for 3D models.

5.2 Applications

The technique can support several applications. The one is video viewing as used in the experiment. In this application, the user can watch a larger video than is possible with only the smartphone. Regions of interest can be carefully checked in higher definition by placing the smartphone there. This application takes advantages of the two devices and eases their disadvantages.

Two applications were implemented to explore the proposal's design space. The first was the creation of a fish game, see Figure 5 (A), which incorporates user's touch input on the smartphone. Several fish, including two sharks, are swimming, and the sharks glow red and disappear at random. The smartphone emulates a camera, and the user is required to take a picture of a shark by tapping the smartphone when it is red. When the shark is captured at the center of the picture, the user gets a high score. The user loses points if the shark is not in the picture or it is white. To implement this game, the predefined hand posture needs to change as the user holds the smartphone with the thumb and middle finger to allow the index finger to tap the smartphone.

The second one is a cross-sectional display for 3D models, see Figure 5 (B). This uses the smartphone's ability to move in 3D space. This application displays a cross-sectional view of a 3D model that changes according to the depth distance of the smartphone.

The user study participants tested these two applications. User comments included, for example, "I used AR glasses to see sharks in my field of vision and followed them with my smartphone," for the first one and "it is good to be able to view the entire 3D model with AR glasses while looking at the cross-sectional view on the smartphone."

6 CONCLUSION

A screen augmentation technique that uses AR glasses and a smartphone is proposed. It does not use external sensors and instead uses a robust built-in hand tracking function. To handle the inevitable tracking errors, the proposal implements a method to make the resulting gaps less perceptible. The tracking accuracy was measured and it was confirmed the technique achieves seamless augmentation. Two applications were implemented to explore the design space using the proposed technique.

REFERENCES

- [1] Sunyoung Bang and Woontack Woo. 2023. Enhancing the Reading Experience on AR HMDs by Using Smartphones as Assistive Displays. In *2023 IEEE Conference Virtual Reality and 3D User Interfaces (VR)*. Shanghai, China, 378–386. <https://doi.org/10.1109/VR55154.2023.00053>

- [2] Eugénie Brasier, Emmanuel Pietriga, and Caroline Appert. 2021. AR-Enhanced Widgets for Smartphone-Centric Interaction. In *Proceedings of the 23rd International Conference on Mobile Human-Computer Interaction* (Toulouse & Virtual, France) (*MobileHCI '21*). Association for Computing Machinery, New York, NY, USA, Article 32, 12 pages. <https://doi.org/10.1145/3447526.3472019>
- [3] Cesar Cadena, Luca Carlone, Henry Carrillo, Yasir Latif, Davide Scaramuzza, José Neira, Ian Reid, and John J. Leonard. 2016. Past, Present, and Future of Simultaneous Localization and Mapping: Toward the Robust-Perception Age. *IEEE Transactions on Robotics* 32, 6 (2016), 1309–1332. <https://doi.org/10.1109/TRO.2016.2624754>
- [4] Neil Chulpongsatorn, Wesley Willett, and Ryo Suzuki. 2023. HoloTouch: Interacting with Mixed Reality Visualizations Through Smartphone Proxies. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (*CHI EA '23*). Association for Computing Machinery, New York, NY, USA, Article 156, 8 pages. <https://doi.org/10.1145/3544549.3585738>
- [5] Carolina Cruz-Neira, Daniel J. Sandin, Thomas A. DeFanti, Robert V. Kenyon, and John C. Hart. 1992. The CAVE: Audio Visual Experience Automatic Virtual Environment. *Commun. ACM* 35, 6 (jun 1992), 64–72. <https://doi.org/10.1145/129888.129892>
- [6] Francesco Riccardo Di Gioia, Eugénie Brasier, Emmanuel Pietriga, and Caroline Appert. 2022. Investigating the Use of AR Glasses for Content Annotation on Mobile Devices. *Proc. ACM Hum.-Comput. Interact.* 6, ISS, Article 574 (nov 2022), 18 pages. <https://doi.org/10.1145/3567727>
- [7] Jens Grubert, Matthias Heinisch, Aaron Quigley, and Dieter Schmalstieg. 2015. MultiFi: Multi Fidelity Interaction with Displays On and Around the Body. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems* (Seoul, Republic of Korea) (*CHI '15*). Association for Computing Machinery, New York, NY, USA, 3933–3942. <https://doi.org/10.1145/2702123.2702331>
- [8] Sebastian Hubenschmid, Johannes Zagermann, Daniel Leicht, Harald Reiterer, and Tiare Feuchtnner. 2023. ARound the Smartphone: Investigating the Effects of Virtually-Extended Display Size on Spatial Memory. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (*CHI '23*). Association for Computing Machinery, New York, NY, USA, Article 527, 15 pages. <https://doi.org/10.1145/3544548.3581438>
- [9] Brett Jones, Rajinder Sodhi, Michael Murdock, Ravish Mehra, Hrvoje Benko, Andrew Wilson, Eyal Ofek, Blair MacIntyre, Nikunj Raghuvanshi, and Lior Shapira. 2014. RoomAlive: Magical Experiences Enabled by Scalable, Adaptive Projector-Camera Units. In *Proceedings of the 27th Annual ACM Symposium on User Interface Software and Technology* (Honolulu, Hawaii, USA) (*UIST '14*). Association for Computing Machinery, New York, NY, USA, 637–644. <https://doi.org/10.1145/2642918.2647383>
- [10] Brett R. Jones, Hrvoje Benko, Eyal Ofek, and Andrew D. Wilson. 2013. IllumiRoom: Peripheral Projected Illusions for Interactive Experiences. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Paris, France) (*CHI '13*). Association for Computing Machinery, New York, NY, USA, 869–878. <https://doi.org/10.1145/2470654.2466112>
- [11] Hiroyuki Kawakita and Toshio Nakagawa. 2014. Augmented TV: An augmented reality system for TV programs beyond the TV screen. In *2014 International Conference on Multimedia Computing and Systems (ICMCS)*. Marrakesh, Morocco, 955–960. <https://doi.org/10.1109/ICMCS.2014.6911158>
- [12] Ricardo Langner, Marc Satkowski, Wolfgang Büschel, and Raimund Dachselt. 2021. MARVIS: Combining Mobile Devices and Augmented Reality for Visual Data Analysis. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (*CHI '21*). Association for Computing Machinery, New York, NY, USA, Article 468, 17 pages. <https://doi.org/10.1145/3411764.3445593>
- [13] Joon Hyub Lee, Taegyu Jin, Sang-Hyun Lee, Seung-Jun Lee, and Seok-Hyung Bae. 2023. Stereoscopic Viewing and Monoscopic Touching: Selecting Distant Objects in VR Through a Mobile Device. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology* (San Francisco, CA, USA) (*UIST '23*). Association for Computing Machinery, New York, NY, USA, Article 28, 7 pages. <https://doi.org/10.1145/3586183.3606809>
- [14] Andrew MacQuarrie and Anthony Steed. 2017. Cinematic virtual reality: Evaluating the effect of display type on the viewing experience for panoramic video. In *2017 IEEE Virtual Reality (VR)*. IEEE Computer Society, Los Angeles, USA, 45–54. <https://doi.org/10.1109/VR.2017.7892230>
- [15] Peter Mohr, Markus Tatzgern, Tobias Langlotz, Andreas Lang, Dieter Schmalstieg, and Denis Kalkofen. 2019. TrackCap: Enabling Smartphones for 3D Interaction on Mobile Head-Mounted Displays. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (*CHI '19*). Association for Computing Machinery, New York, NY, USA, 1–11. <https://doi.org/10.1145/3290605.3300815>
- [16] Erwan Normand and Michael J. McGuffin. 2018. Enlarging a Smartphone with AR to Create a Handheld VESAD (Virtually Extended Screen-Aligned Display). In *2018 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. Munich, Germany, 123–133. <https://doi.org/10.1109/ISMAR.2018.00043>
- [17] Carolin Reichherzer, Jack Fraser, Damien Constantine Rompapas, and Mark Billinghurst. 2021. SecondSight: A Framework for Cross-Device Augmented Reality Interfaces. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (*CHI EA '21*). Association for Computing Machinery, New York, NY, USA, Article 234, 6 pages. <https://doi.org/10.1145/3411763.3451839>
- [18] Pejman Saeghe, Sarah Clinch, Bruce Weir, Maxine Glancy, Vinoba Vinayagamoorthy, Ollie Pattinson, Stephen Robert Pettifer, and Robert Stevens. 2019. Augmenting Television With Augmented Reality. In *Proceedings of the 2019 ACM International Conference on Interactive Experiences for TV and Online Video* (Salford (Manchester), United Kingdom) (*TVX '19*). Association for Computing Machinery, New York, NY, USA, 255–261. <https://doi.org/10.1145/3317697.3325129>
- [19] Houssein Saidi, Emmanuel Dubois, and Marcos Serrano. 2021. HoloBar: Rapid Command Execution for Head-Worn AR Exploiting Around the Field-of-View Interaction. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (*CHI '21*). Association for Computing Machinery, New York, NY, USA, Article 745, 17 pages. <https://doi.org/10.1145/3411764.3445255>
- [20] Arda Ege Unlu and Robert Xiao. 2021. PAIR: Phone as an Augmented Immersive Reality Controller. In *Proceedings of the 27th ACM Symposium on Virtual Reality Software and Technology* (Osaka, Japan) (*VRST '21*). Association for Computing Machinery, New York, NY, USA, Article 27, 6 pages. <https://doi.org/10.1145/3489849.3489878>
- [21] Vinoba Vinayagamoorthy, Maxine Glancy, Christoph Ziegler, and Richard Schäfer. 2019. Personalising the TV Experience Using Augmented Reality: An Exploratory Study on Delivering Synchronised Sign Language Interpretation. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (*CHI '19*). Association for Computing Machinery, New York, NY, USA, 1–12. <https://doi.org/10.1145/3290605.3300762>
- [22] Fengyuan Zhu and Tovi Grossman. 2020. BISHARE: Exploring Bidirectional Interactions Between Smartphones and Head-Mounted Augmented Reality. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (*CHI '20*). Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3313831.3376233>